# OASIS Agentic Pipeline

Effectiveness & Data Analysis Report

Automated evaluation report generated 2026-06-17.

- RESEARCH USE ONLY - NOT a medical device; not for clinical or diagnostic use.
- MRI scans evaluated: 160 from subject-disjoint HELD-OUT patients (no subject overlap with training) across 4 classes
- Clinical cohort: 100 subjects | Longitudinal records: 373
- Acceleration: ONNX Runtime QNN (Snapdragon NPU) → DirectML → CPU fallback
- Language reasoning: local Ollama / cost-tiered Claude (hybrid)

Hybrid edge-cloud, Snapdragon-NPU-optimized, zero paid-API multi-agent system for Alzheimer's disease screening on the OASIS datasets.

# 1. Executive Summary

| 43.8% | 43.8% | 86.7% | 100.0% |
| --- | --- | --- | --- |
| Overall accuracy | Balanced acc. | Screening sensitivity | Guardrail acc. |

RESEARCH USE ONLY. This is a research prototype and screening decision-support demonstrator - NOT a medical device, and not cleared by any regulator (FDA/CE). It must not be used for clinical diagnosis or patient management.

Validation note: metrics below are computed on a SUBJECT-DISJOINT held-out test set (no patient appears in both training and test) - the honest measure of generalization. The bundled OASIS-1 slice set is patient-poor in some classes (e.g. Moderate Dementia has only ~2 subjects total), so small-class metrics are noisy and indicative only; clinical-grade claims require a large multi-site cohort (OASIS-3 / ADNI).

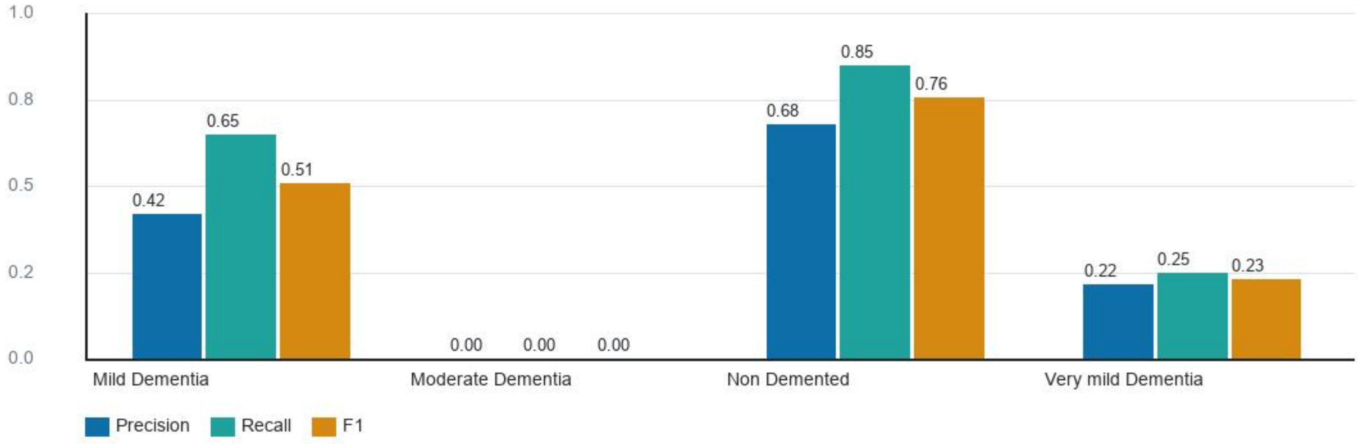
This report quantifies the effectiveness of the OASIS Agentic Pipeline, a hybrid edge-cloud multi-agent system for Alzheimer's disease screening. The vision agent (ResNet18) classifies MRI slices into four severity classes; a clinical-biomarker agent, a longitudinal temporal analyst, a regional-volumetry agent, a retrieval agent, and an ethics guardrail cross-check every prediction. A local Ollama-backed reasoner produces the final grounded narrative — entirely on free, on-device models.

- Cohen's kappa (chance-corrected agreement): 0.250.
- Mean model confidence on evaluated slices: 65.0%.
- Ethics guardrail correctly adjudicated 8/8 adversarial safety cases.

# 2. Vision Agent Classification Performance

Evaluated on 160 slices from SUBJECT-DISJOINT held-out patients (no subject overlap with training), using the training-time validation transform (grayscale, 224x224, normalized). Held-out test subjects per class: Mild Dementia=4, Moderate Dementia=1, Non Demented=53, Very mild Dementia=12. A class with very few held-out subjects yields a statistically meaningless per-class metric.

### Per-class precision / recall / F1




| Class | Precision | Recall | F1 | Support |
|---|---|---|---|---|
| Mild Dementia | 0.42 | 0.65 | 0.51 | 40 |
| Moderate Dementia | 0.00 | 0.00 | 0.00 | 40 |
| Non Demented | 0.68 | 0.85 | 0.76 | 40 |
| Very mild Dementia | 0.22 | 0.25 | 0.23 | 40 |

## 2.1 Confusion Matrix

### Confusion matrix (true vs predicted)

| | Mild Dementia | Moderate Dementia | Non Demented | Very mild Dementia |
|---|---|---|---|---|
| Mild Dementia | 26<br>65% | 2<br>5% | 1<br>2% | 11<br>28% |
| Moderate Dementia | 16<br>40% | 0<br>0% | 5<br>12% | 19<br>48% |
| Non Demented | 0<br>0% | 0<br>0% | 34<br>85% | 6<br>15% |
| Very mild Dementia | 20<br>50% | 0<br>0% | 10<br>25% | 10<br>25% |

Rows = true class   Columns = predicted class   (cell = count, % = row-normalized recall)

## 2.2 Clinical screening view

Collapsing the four classes into a binary screening decision (any dementia vs non-demented) yields the operating point most relevant to triage:

- Sensitivity (impaired correctly flagged): 86.7%
- Specificity (healthy correctly cleared): 85.0%

# 3. Cohort & Longitudinal Data Analysis

Cross-sectional cohort: 100 subjects. Age 74.8±7.3 yr; MMSE 24.9±3.3; mean nWBV 0.729.

- Sex distribution: F=56, M=44

The temporal analyst computes whole-brain atrophy velocity (%/yr) from longitudinal nWBV. Group separation validates the longitudinal agent:

### Mean atrophy velocity by diagnostic group (%/yr)

# 4. Ethics Guardrail Effectiveness

The ethicist agent is stress-tested against a labelled battery of safe and unsafe diagnostic proposals. It must flag unsafe/contradictory cases and pass consistent ones.

**100%**
Battery accuracy

**8**
Cases tested

| Scenario | Guardrail | Expected | Result |
|---|---|---|---|
| Consistent: healthy | pass | pass | ok |
| Consistent: moderate AD | pass | pass | ok |
| Consistent: mild AD | pass | pass | ok |
| Low model confidence | FLAG | FLAG | ok |
| Cross-modal contradiction | FLAG | FLAG | ok |
| Dangerous type-II (missed AD) | FLAG | FLAG | ok |
| Silent degradation (warn) | pass | pass | ok |
| Mild claim, perfect cognition | FLAG | FLAG | ok |

# 5. Regional Volumetry (FreeSurfer aseg)

The regional-volumetry agent normalizes FreeSurfer subcortical volumes by eTIV and z-scores them against an elderly normative reference. Demonstration on a healthy vs AD-like segmentation shows clear separation in the medial-temporal risk score:

| Subject | Hippocampus z | Ventricle z | MTA risk | Stage |
|---|---|---|---|---|
| Healthy reference | +0.05 | +0.09 | 0.04 | Within normal limits |
| Atrophic (AD-like) | -2.24 | -2.09 | 2.12 | Severe medial-temporal atrophy (MTA-lik |

### Medial-temporal-atrophy risk score

# 6. Edge Acceleration Benchmark (Snapdragon NPU)

**96.3 MB**
FP32 model

**12.1 MB**
INT8 model

**87%**
Size reduction

### Per-inference latency by execution provider (ms, lower is better)


| Execution provider | Latency | Status |
| --- | --- | --- |
| CPU | 69.7 ms | available |
| PyTorch FP32 (reference) | 37.6 ms | baseline |

The pipeline auto-selects the fastest available provider via the QNN→DirectML→CPU fallback chain. On Snapdragon X hardware the QNN row maps to the Hexagon NPU; on other machines it gracefully degrades to DirectML or CPU.

# 7. Methodology & Limitations

- Vision metrics use a SUBJECT-DISJOINT (patient-grouped) split - no subject appears in both training and test (src/pipeline/data_split.py). This is the honest measure of generalization; image-level splits inflate accuracy because there are ~240 near-identical slices per subject. Evaluated on CPU with training-consistent normalization (seed=42).
- Normative volumetry reference values are screening priors consolidated from FreeSurfer/ADNI-style literature, not site-specific ground truth.
- Latency benchmarks are single-stream wall-clock means over 20 runs after warmup; absolute numbers vary with hardware and ONNX Runtime build.
- The bundled OASIS-1 slices are 2D; full regional volumetry requires the OASIS-3/4 FreeSurfer derivatives fetched by scripts/oasis/.
- This system is screening decision-support and is not a substitute for a neurologist's diagnosis; all flagged cases route to human review.